# Noise-Robust Bird Vocalization Classification for Low-Resource Scenarios: A Multi-Paradigm Comparative Study


1st Jianan Zhang
*Department of Computer Science*
*Stevens Institute of Technology*
Beijing, China
email: jzhang112@stevens.edu

2nd Jincheng Chen
*Department of Computer Science*
*Stevens Institute of Technology*
Guangzhou, China
email: jchen174@stevens.edu

3rd Wenjuan Huang
*Department of Computer Science*
*Stevens Institute of Technology*
Zhejiang, China
email: whuang32@stevens.edu



***Abstract*—Automated bird vocalization classification from passive acoustic monitoring is critical for biodiversity conservation, yet field recordings present two compounding challenges: environmental noise that masks subtle species-specific features, and extreme long-tail distributions where rare species have minimal training samples. Existing solutions such as BirdNET and Google Perch lack standardized noise robustness evaluation and do not provide cross-paradigm comparisons incorporating traditional machine learning baselines. This work presents a systematic multi-paradigm comparative study on a unified dataset of 129,834 recordings spanning 1,126 bird species from BirdCLEF 2021–2026. We evaluate three approaches: (1) frozen YAMNet with a Dense head on 3072-dimensional clip-level pooled embeddings (mean, max, and std pooling over 1024-dimensional frame-level YAMNet outputs every 0.48 s; encoder not fine-tuned); (2) LightGBM on the same 3072-dimensional pooled features—a cascade decoupling neural feature extraction from tree-based classification, achieving faster inference (7.7 ms/s) than BirdNET (∼10.0 ms/s) with modular flexibility; and (3) FastAI ConvNeXt-Small on mel-spectrograms. All share identical 5-fold CV splits, the same noise-evaluation protocol, CSV `loss_class_weight`, and targeted audio augmentation for low-resource species (<15 samples). Noise robustness is evaluated under Gaussian white noise at 5 dB, 0 dB, and −5 dB SNR. Our primary contribution is a reproducible evaluation framework with common data splits and noise testing for direct quantitative comparison across heterogeneous ML paradigms—a capability absent from existing bioacoustic tools.**


## I. Introduction

Automated bird vocalization classification is essential for large-scale biodiversity monitoring, yet field recordings pose dual challenges: environmental noise masks discriminative spectral features, and species distributions follow an extreme long-tail pattern where rare species have minimal training data [1]–[3]. These compounding factors severely degrade classifier performance, particularly for the very species most in need of conservation monitoring.

Existing off-the-shelf solutions such as BirdNET [1] and Google Perch [2] demonstrate significant limitations under these conditions. BirdNET's accuracy degrades markedly when SNR falls below 3 dB [3], and its recall varies drastically across regions—from PR AUC 0.16–0.23 in North America and Europe to 0.03–0.04 in Africa and Asia [3]. Perch achieves only ∼60% accuracy on BirdCLEF+ 2025 data containing species outside its training set [2]. Critically, neither provides standardized, quantifiable noise robustness evaluation nor cross-paradigm comparisons that include traditional ML baselines.

This project addresses these gaps through three pipelines on a unified 129,834-sample dataset with controlled noise evaluation: (1) frozen YAMNet (MobileNetV1, AudioSet [4]) with a Dense head on 3072-dimensional clip-level pooled embeddings (aggregated from 1024-dimensional frame-level outputs); (2) LightGBM on the same 3072-dimensional pooled features—a two-stage cascade decoupling neural feature extraction from tree-based classification, enabling GPU-free training and modular experimentation; and (3) FastAI ConvNeXt-Small on mel-spectrograms. All share identical 5-fold CV, the same SNR evaluation protocol, CSV `loss_class_weight`, and targeted audio augmentation for species with fewer than 15 training samples. Evaluation uses clean audio plus Gaussian noise at 5 dB, 0 dB, and −5 dB SNR. While we use Gaussian white noise as a controlled, reproducible baseline, qualitatively different field noise (wind, rain, insect interference) is discussed in Section V.

Our contributions are threefold: (i) a reproducible cross-paradigm evaluation framework with common data splits and standardized noise testing; (ii) empirical insights into trade-offs between frozen embedding classifiers, embedding-based gradient boosting, and spectrogram-based classification—with emphasis on the YAMNet+LightGBM cascade as a computationally efficient alternative to monolithic systems; and (iii) a practical framework combining shared CSV-based class weighting and targeted low-resource augmentation with standardized evaluation for low-resource bioacoustic classification.

## II. Related Work

### A. BirdNET and Google Perch: Off-the-Shelf Solutions

BirdNET [1] (Cornell Lab/Chemnitz University) uses an EfficientNetB0-like backbone with 1024-dimensional embeddings, classifying 48 kHz audio into 3,337 classes [7]. Despite its widespread adoption, it exhibits: (1) severe performance degradation below 3 dB SNR [3]; (2) highly heterogeneous recall across regions (0.24–0.52) and species rarity levels [3]; and (3) label noise and dataset imbalances as primary error sources [8]. Google Perch [2] (EfficientNet-B1) achieves state-of-the-art on BirdSET and BEANS benchmarks [9], [10] but only $\sim$60% accuracy on BirdCLEF+ 2025 data with out-of-training-set species. Critically, neither provides standardized controllable noise testing pipelines.

### B. BirdCLEF Competitions: Persistent Challenges

The BirdCLEF series (2021–2026) consistently benchmarks core challenges: long-tailed distributions, weak labeling (recording-level vs. segment-level), and domain shift between Xeno-Canto training and PAM test soundscapes [5], [11]. BirdCLEF+ 2025 saw foundation models like BEATs achieve 0.913 F1 with pseudo-labeling [12]. These competitions underscore that no integrated solution spans data, model, and evaluation dimensions for low-resource noisy environments.

### C. Noise Robustness and Low-Resource Classification

Environmental noise remains a formidable obstacle. Noise reduction preprocessing can help but occasionally degrades performance by removing informative features [13]. Data augmentation with real environmental noise under varying SNR has proven effective [5], [14], and generative approaches (AC-GANs, DDPMs) show promise [15]. For low-resource species, transfer learning is essential: avian-specific embeddings outperform general audio embeddings [16], and AudioSet pretraining followed by fine-tuning boosts low-resource performance [17], [16]. Few-shot approaches (Siamese networks) target rare species absent from BirdNET/Perch [18]. However, existing studies rarely provide systematic cross-paradigm comparisons on identical datasets.

### D. YAMNet and Audio Transfer Learning

YAMNet [4], [17] (MobileNetV1, AudioSet-trained) outputs 1024-dimensional frame-level embeddings at 0.48-second intervals. In our pipelines, these frames are aggregated by mean, maximum, and standard deviation into 3072-dimensional clip-level pooled features for downstream classifiers. It has been adapted for speech recognition, urban noise, and industrial monitoring [19]. While domain-specific models (Perch, BirdNET) outperform general-purpose ones on complex calls [16], [20], YAMNet provides a computationally accessible baseline for raw audio classification.

### E. Research Gap Summary

The literature reveals six critical gaps: (1) no unified framework comparing tabular ML, image CNN, and raw-audio transfer learning on identical datasets with standardized noise evaluation; (2) off-the-shelf tools lack quantifiable controllable noise pipelines; (3) no systematic cross-paradigm comparison for rare species; (4) no reproducible controlled-variable experiment separating clean baseline from gradient noise; (5) BirdCLEF solutions rarely integrate data, model, and evaluation dimensions; and (6) no systematic evaluation of hybrid pipelines combining deep embeddings with lightweight tree classifiers—which can outperform monolithic systems while being faster and more deployable. Our YAMNet+LightGBM cascade directly addresses gap (6), achieving $\sim$7.7 ms/s embedding latency (faster than BirdNET's $\sim$10.0 ms/s) with modular flexibility.

TABLE I: Final Dataset Split (V2.0)

| Split | Records | Species | Use |
|---|---|---|---|
| Cleaned population | 144,250 | 1,126 | Source |
| Training pool | 129,834 | 1,126 | Development & CV |
| CV validation (per fold) | $\sim$25,967 | 1,126 | In-fold validation (5-fold SNR-stratified CV) |
| Fixed holdout test | 14,416 | 1,093 | Final evaluation |

TABLE II: SNR Tier Distribution Across Train/Test Splits

| SNR Tier | Training | Test |
|---|---|---|
| High ($\geq$4.0) | 86,731 | 9,631 |
| Medium (2.0–3.9) | 41,011 | 4,553 |
| Heavy (<2.0) | 2,092 | 232 |

## III. Our Solution

### A. Dataset

*1) Data Source and Collection:* Compiled from BirdCLEF 2021–2026, aggregating Xeno-Canto and iNaturalist recordings [5], [6]. Raw merge of six annual CSVs: 183,239 records. Field-level deduplication (zero-information-loss merge) yielded 167,308 unique recordings.

*2) Data Cleaning and Quality Control:* Two filters applied:

1) Remove numeric-only `primary_label` (non-avian taxa: Amphibia, Insecta, Mammalia, Reptilia)—excludes 101 non-bird species;
2) Discard rating $\leq$ 0.5 (unrated/extremely low-quality).

Final cleaned population: **144,250 recordings across 1,126 bird species**.

SNR proxy tier derived from rating: High ($\geq$4.0), Medium (2.0–3.9), Heavy (<2.0). Split: 90/10 with SNR stratification (seed=42). Species with <5 recordings retained entirely in training. Tables I–II summarize the split.

5-fold SNR-stratified CV: each fold $\sim$103,853 training, $\sim$25,967 validation. No filename overlap between train/test or train/val within folds. This represents a **10$\times$ scale-up** from the original proposal (12,400 $\rightarrow$ 129,834), enabling robust long-tail evaluation.

*3) Long-Tail Distribution and Mitigation:* Species distribution: median $\sim$72 training samples/species, minimum 1, head class 1,300. **Shared long-tail mitigation (all three pipelines)**:

1) **Balanced sampling**: $w_{\text{sampler}} = 1/N_c$, ensuring equal per-species batch probability (exported for LightGBM; not used in Dense-head training).
2) **Mild class-weighted loss**:

$$w_{\text{loss}}(c) = \text{clip}\left(\frac{1/\sqrt{N_c}}{\mu_w},\, 0.25,\, 4.00\right) \quad (1)$$

Inverse-square-root is milder than $1/N_c$; clipping ensures stability. Species with fewer than 15 training samples are treated as low-resource tail classes, consistent with BirdCLEF findings that such species exhibit near-random performance [11].

3) **Targeted audio augmentation**: 159 species (14.1%) with <15 training examples receive four on-the-fly waveform transformations implemented in `audio_augmentation.py`: time stretching (0.85×–1.10×), pitch shifting (±1–±2 semitones), additive Gaussian noise (5–15 dB SNR), and volume scaling (0.70×–1.30×), capped at 16 variants per recording and 50 new variants per species, with a per-species target of 15 samples. This augmentation is enabled in all three reported pipelines: YAMNet applies `expand_with_augmentation` before embedding extraction and Dense-head training; LightGBM trains on 3072-dimensional pooled embeddings computed from the same augmented training folds; FastAI generates augmented mel-spectrogram PNGs via `augmentation_glue.py` before ConvNeXt training.

*4) Evaluation Protocol:* Fixed test set (14,416) reserved for final evaluation only. Metrics:

1) clean Top-1 accuracy;
2) noise robustness at 5 dB, 0 dB, −5 dB SNR (Gaussian white noise, fixed seed=42, peak-normalized);
3) per-SNR-tier stratified accuracy;
4) inference latency (ms/sample) and peak GPU memory.

### B. Machine Learning Algorithms

*1) Frozen YAMNet Embeddings with Dense Head:* **Rationale.** MobileNetV1 pre-trained on AudioSet's 521 categories [4] provides general-purpose acoustic features transferable to bird vocalizations. AudioSet includes environmental noise [17], giving inherent noise-robust representations. We freeze the encoder and train only a lightweight head on pooled embeddings [16].

**Design.** 16 kHz mono, 5-second clips (center-crop or zero-pad), peak-normalized. (1) Frozen YAMNet encoder (`trainable=False`) outputs 1024-dimensional frame-level embeddings every 0.48 s; mean, max, and std pooling across frames → 3072-dimensional clip-level feature vector. (2) Dense head: Dense(256, ReLU) → Dropout(0.3) → Dense(1126, Softmax). Adam (lr $= 10^{-3}$), up to 50 epochs, early stopping (patience=8), `loss_class_weight` as sample weights. Low-resource species (<15 samples) are expanded via `expand_with_augmentation` before embedding extraction. Embeddings are cached once and exported per fold for LightGBM. Noise evaluation re-computes clip-level pooled embeddings on noisy test audio.

*2) LightGBM: Gradient-Boosted Decision Trees on YAMNet Embeddings:* **Rationale.** Gradient-boosted trees complement deep learning: no GPU required, fast CPU training, native feature importance, robust to hyperparameters. However, trees cannot operate on raw audio. Our **cascade architecture** bridges this: YAMNet's frozen encoder extracts embeddings; LightGBM classifies. This decoupling offers three advantages: (i) embedding extraction runs once per file; (ii) classification is GPU-free, trainable in minutes with class-weighted loss; (iii) modular—same embeddings reusable for different classifiers.

**Audio Preprocessing.** 16 kHz mono OGG loaded directly via librosa (no WAV conversion). Fixed 5.0 seconds (zero-pad if shorter; center-crop if longer), peak-normalized to $[-1, 1]$. **No noise reduction**—evaluation requires measuring robustness against controlled additive noise.

**Feature Extraction.** The frozen YAMNet encoder produces 1024-dimensional frame-level embeddings at 0.48 s intervals (∼10 frames per 5 s clip). These are pooled by **mean, maximum, and standard deviation** into a **3072-dimensional clip-level feature vector**—not the raw 1024-d frame outputs. This captures both average spectral profile and temporal variability—critical for dynamic vocalization patterns. Compared to handcrafted features (MFCC statistics, spectral centroid), YAMNet pooled features preserve deeper semantic information; compared to flattened spectrograms (∼64,000 dims), 3072 dims makes tree learning tractable.

**Model Design.** LightGBM: 1,500 rounds with early stopping (patience=50), learning rate 0.05, max_depth=6, num_leaves=31, L1/L2 regularization ($\alpha = 0.1$, $\lambda = 0.1$), `loss_class_weight` as sample weights. CPU inference only. Outputs 1,126-dimensional raw class scores under the multiclass objective (margin logits, not native softmax). For probability-based metrics, scores are normalized via softmax at evaluation time.

**Methodological value:** Establishes performance floor for CPU-only tools with high-quality embeddings—quantifying accuracy-per-compute trade-offs against deep neural classifiers on the same embeddings.

*3) FastAI ConvNeXt: Mel-Spectrogram Image Classification:* **Rationale.** Mel-spectrograms expose harmonic structures, formant patterns, and temporal modulations. ConvNeXt-Small provides strong performance with ImageNet transfer learning [14], [11]. Long-tail imbalance is handled with the same CSV `loss_class_weight` as YAMNet/LightGBM, supplemented by spectrogram-level MixUp and RandomErasing. Low-resource waveform augmentation is applied upstream via `augmentation_glue.py` before PNG caching.

**Design.** 16 kHz mono; highest-RMS 5 s segment; 128 mel bands (`n_fft=1024`, hop 512, 500–8000 Hz) → Viridis RGB images at 224×224 (cached offline, including augmented low-resource samples). ConvNeXt-Small, FP16, CSV `loss_class_weight` as sample weights, MixUp ($\alpha = 0.4$), RandomErasing ($p = 0.4$). `fine_tune`: 3 frozen epochs + 12 full-network epochs (`base_lr`=$2\times10^{-3}$). Metrics: Top-1 and macro-F1.

### C. Implementation Details

All pipelines share:

- Identical 5-fold CV splits (SNR-stratified)
- Identical noise evaluation: Gaussian white noise at 5/0/−5 dB, fixed seed=42
- Same metrics: Top-1, Top-5, macro-F1, balanced accuracy, stratified by SNR tier and class frequency group
- Inference measurement: 5 warm-up + 50 timed iterations; GPU memory profiling for YAMNet/FastAI; CPU-only for LightGBM

- Shared long-tail controls: all pipelines use CSV `loss_class_weight` and targeted audio augmentation for species with $<15$ samples; FastAI additionally applies MixUp and RandomErasing at the spectrogram level
- Embedding terminology: YAMNet outputs 1024-d frame-level embeddings; Dense/LightGBM consume 3072-d clip-level pooled features (mean/max/std)

## IV. Comparison

### A. Experimental Results

[To be completed after model training and evaluation. Will present: (1) comparative Top-1, Top-5, macro-F1 on clean test audio; (2) noise robustness degradation curves (5 dB, 0 dB, $-5$ dB); (3) per-SNR-tier and per-class-frequency-group stratified performance; (4) inference latency and GPU memory benchmarks.]

### B. Comparison with Existing Solutions

[To be completed. Will compare best-performing model against BirdNET and Perch baselines on the same test set, analyzing trade-offs between transfer learning, YAMNet embedding-based gradient boosting, and CNN image classification.]

## V. Future Directions

1) **Multi-label classification**: Incorporate `secondary_labels` for partially labeled co-occurrences, reducing label noise and improving tail-class performance.
2) **Attention-based embedding aggregation**: Replace simple mean/max/std pooling with learnable attention to preserve temporally localized discriminative patterns.
3) **Extended noise conditions**: Evaluate against wind, rain, and insect noise—qualitatively different from Gaussian.
4) **Model distillation**: Distill YAMNet+LightGBM cascade into a single lightweight edge-deployable model.
5) **Active learning for rare species**: Identify most informative unlabeled recordings for human annotation, maximizing labeling efficiency for species with $<5$ samples.

## VI. Conclusion

[To be completed after experimental results are obtained. This section will summarize: whether the noise-robust bird vocalization classification problem is adequately addressed by our multi-paradigm approach; which algorithm(s) demonstrate superior performance under specific conditions (clean vs. noisy, head vs. tail classes); which shared techniques (CSV `loss_class_weight`, targeted low-resource augmentation, frozen AudioSet embeddings vs. ConvNeXt spectrogram modeling) contribute most to performance gains; and broader implications for low-resource bioacoustic classification system design.]


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